[Assignment - 4]

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section : 05

course code : CSE 330

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Ans to the Q: No: 1

a)

$$x_1 + 6x_2 + 2x_3 = 10$$

$$3x_1 + 2x_2 + x_3 = 6$$

$$4x_1 + 5x_2 + 2x_3 = 9$$

$$\therefore A = \begin{bmatrix} 1 & 6 & 2 \\ 3 & 2 & 1 \\ 4 & 5 & 2 \end{bmatrix}, \quad x = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}, \quad b = \begin{bmatrix} 10 \\ 6 \\ 9 \end{bmatrix}$$

such that, $Ax = b$

b)

$$F^{(1)} = \begin{bmatrix} 1 & 0 & 0 \\ -3 & 1 & 0 \\ -4 & 0 & 1 \end{bmatrix}$$

$m_{21} = 3$

$m_{31} = 4$

$$A^{(2)} = F^{(1)} \times A$$

$$= \begin{bmatrix} 1 & 0 & 0 \\ -3 & 1 & 0 \\ -4 & 0 & 1 \end{bmatrix} \times \begin{bmatrix} 1 & 6 & 2 \\ 3 & 2 & 1 \\ 4 & 5 & 2 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 6 & 2 \\ 0 & -16 & -5 \\ 0 & -19 & -6 \end{bmatrix}$$

$$m_{32} = \frac{-19}{-16} = \frac{19}{16}$$

$$F^{(2)} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & -\frac{19}{16} & 1 \end{bmatrix}$$

c)

$$L = \begin{bmatrix} 1 & 0 & 0 \\ m_{21} & 1 & 0 \\ m_{31} & m_{32} & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 3 & 1 & 0 \\ 4 & \frac{19}{16} & 1 \end{bmatrix}$$

d)

$Ly = b$

$$\Rightarrow \begin{bmatrix} 1 & 0 & 0 \\ 3 & 1 & 0 \\ 4 & \frac{19}{16} & 1 \end{bmatrix} \times \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix} = \begin{bmatrix} 10 \\ 6 \\ 9 \end{bmatrix}$$

Forward substitution:

$$y_1 = 10$$

$$3y_1 + y_2 = 6$$

$$\Rightarrow 3 \times 10 + y_2 = 6$$

$$\Rightarrow y_2 = -24$$

$$4y_1 + \frac{19}{16} y_2 + y_3 = 9$$

$$\Rightarrow 4 \times 10 + \frac{19}{16} \times (-24) + \cancel{(-24)} = y_3 = 9$$

$$\therefore y_3 = -\frac{5}{2}$$

$$\therefore y = \begin{bmatrix} 10 \\ -24 \\ -\frac{5}{2} \end{bmatrix}$$

(Ans)

$Ux = y$

$$\Rightarrow \begin{bmatrix} 1 & 6 & 2 \\ 0 & -16 & -5 \\ 0 & 0 & -\frac{1}{16} \end{bmatrix} \times \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 10 \\ -24 \\ -\frac{5}{2} \end{bmatrix}$$

Backward substitution:

$$-\frac{1}{16} \times x_3 = -\frac{5}{2}$$

$$\Rightarrow x_3 = \frac{5}{2} \times 16$$

$$\therefore x_3 = 40$$

$$-16x_2 - 5x_3 = -24$$

$$\Rightarrow -16x_2 = -24 + 5 \times 40$$

$$\Rightarrow x_2 = \frac{176}{-16}$$

$$\therefore x_2 = -11$$

$$x_1 + 6x_2 + 2x_3 = 10$$

$$\Rightarrow x_1 = 10 - 6 \times (-11) - 2(40)$$

$$\therefore x_1 = -4$$

$$\therefore x_1 = -4$$
$$x_2 = -11$$
$$x_3 = 40$$

(Ans)

$$A^{(3)} = F^{(2)} \times A^{(2)}$$

$$= \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & -\frac{19}{16} & 0 \end{bmatrix} \times \begin{bmatrix} 1 & 6 & 2 \\ 0 & -16 & -5 \\ 0 & -19 & -6 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 6 & 2 \\ 0 & -16 & -5 \\ 0 & 0 & -\frac{1}{16} \end{bmatrix}$$

Ans to the question no 2

a) $$6x_2 + 2x_3 = 10$$

$$3x_1 + 2x_2 + x_3 = 6$$

$$4x_1 + 5x_2 + 2x_3 = 9$$

$$A = \begin{bmatrix} 0 & 6 & 2 \\ 3 & 2 & 1 \\ 4 & 5 & 2 \end{bmatrix}, \quad x = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}, \quad b = \begin{bmatrix} 10 \\ 6 \\ 9 \end{bmatrix}$$

such that, $Ax = b$

b) Yes, the matrix A has a pivoting problem because the first pivot element $a_{11} = 0$. This could cause division by zero in Gaussian elimination without row exchange.

c) $$\text{Aug}(A) = \left[\begin{array}{ccc|c} 0 & 6 & 2 & 10 \\ 3 & 2 & 1 & 6 \\ 4 & 5 & 2 & 9 \end{array}\right]$$

since,

$a_{11} = 0$, we are swapping row 1 and row 3:

$$\left[\begin{array}{ccc|c} 4 & 5 & 2 & 9 \\ 3 & 2 & 1 & 6 \\ 0 & 6 & 2 & 10 \end{array}\right]$$

$$m_{21} = \frac{3}{4}$$

$$r_2' = r_2 - \frac{3}{4} r_1$$

$$\therefore \left[\begin{array}{ccc|c} 4 & 5 & 2 & 9 \\ 0 & -\frac{7}{4} & -0.5 & -\frac{3}{4} \\ 0 & 6 & 2 & 10 \end{array}\right]$$

$$m_{32} = \frac{6}{-\frac{7}{4}} = 6 \times -\frac{4}{7} = -\frac{24}{7}$$

$$r_3' = r_3 - \left(-\frac{24}{7}\right) r_2$$

$$\left[\begin{array}{ccc|c} 4 & 5 & 2 & 9 \\ 0 & -\frac{7}{4} & -0.5 & -\frac{3}{4} \\ 0 & 0 & \frac{2}{7} & \frac{52}{7} \end{array}\right]$$

$$\therefore U = \begin{bmatrix} 4 & 5 & 2 \\ 0 & -7/4 & -\frac{1}{2} \\ 0 & 0 & 2/7 \end{bmatrix}$$

d)

$$\left[\begin{array}{ccc|c} 4 & 5 & 2 & 9 \\ 0 & -\frac{7}{4} & -\frac{1}{2} & -\frac{3}{4} \\ 0 & 0 & \frac{2}{7} & \frac{52}{7} \end{array}\right]$$

Backward substitution,

$$\frac{2}{7}x_3 = \frac{52}{7}$$

$$\therefore x_3 = \frac{52}{\cancel{7}} \times \frac{\cancel{7}}{2} = 26$$

$$-\frac{7}{4}x_2 - \frac{1}{2}x_3 = -\frac{3}{4}$$

$$\Rightarrow -\frac{7}{4}x_2 - \frac{1}{2} \times 26 = -\frac{3}{4}$$

$$\therefore x_2 = -7$$

$$4x_1 + 5x_2 + 2x_3 = 9$$

$$\Rightarrow 4 \times x_1 + 5 \times (-7) + 2(26) = 9$$

$$\therefore x_1 = -2$$

$$\therefore x_1 = -2$$
$$x_2 = -7$$
$$x_3 = 26$$